MANAGING SPORT AND LEISURE
2021, VOL. 26, NO. 3, 226–244
https://doi.org/10.1080/23750472.2020.1757492

Routledge
Taylor & Francis Group

Check for updates

The elements of cross-sectoral collaboration between primary care and the sports sector

Satu Parjanen

School of Engineering Science, Lappeenranta-Lahti University of Technology LUT, Lahti, Finland

ABSTRACT

Purpose: The purpose of this study is to define the critical elements of collaboration between primary care and the sports sector to better answer the challenge of physical inactivity. The study also sheds light on how this collaboration could be further developed in the relevant organisations.

Research methods: The data was collected from health care professionals (n=58), physical activity counsellors (n=8) and potential clients of physical activity counselling (n=13) using a questionnaire, observation of workshops, and focus group interviews.

Results and findings: The common objective of the role of physical activity promotion and managerial support create the foundation for a counselling and referral process. At the employee level, attitudes towards physical activity, knowledge about physical activity, and the social skills are emphasised. The physical activity promotion process needs to be continuously evaluated based on objectives formulated by professionals and other stakeholders.

Implications: Managerial actions, such as communication about the essence of physical activity promotion, creation of roles and responsibilities and facilitation of collaboration between professionals enhance collaboration.

Research contribution: There are lack of research on partnership-related processes in the collaboration between the primary care and the sports sector. This paper contributes by presenting a practical case of such collaboration.

ARTICLE HISTORY
Received 18 September 2018
Accepted 15 April 2020

KEYWORDS
Physical activity counselling; sports sector; health professional; physical activity counsellor; cross-sectoral collaboration

Introduction

Physical inactivity levels are rising in many countries, a fact that has major implications for the prevalence of noncommunicable diseases and the general health of the population worldwide (World Health Organization, 2010). There have been changes in people's exercise habits, weight and physical activity. New technology and labour-saving devices are substantially reducing the need for physical activity and, for example, the amount of time that is spent in sedentary behaviours has increased (Thorp et al., 2011). The fact is that people who do not get sufficient exercise to maintain good health, who are overweight or obese and who are in poor physical condition are at a greater risk of getting ill (Booth et al., 2012; Husu et al., 2011; Professional Associations for Physical Activity, 2010). For municipalities this means an increase in social welfare and health care expenditure (Garrett et al., 2004; Hagberg, 2007). Physical inactivity has been connected

CONTACT Satu Parjanen satu.parjanen@lut.fi School of Engineering Science, Lappeenranta-Lahti University of Technology LUT, Mukkulankatu 19, Lahti FI-15210, Finland

© 2020 Informa UK Limited, trading as Taylor & Francis Group

to increased all-cause mortality, lower quality of life, and a higher risk of obesity, diabetes, hypertension, coronary heart disease, osteoporosis, fractures, cancers, mental disorders and an overall greater risk of hospitalisation (Allender et al., 2007; World Health Organization, 2010).

Physical inactivity is a complex challenge for which solutions are not accomplished within one organisation or with the knowledge, experiences and skills of a single profession (Ingerslev, 2016; Parjanen & Hyypiä, 2018). However, collaboration across functional or professional boundaries could be problematic because knowledge is localised, embedded, and invested in collective practice underpinned by shared language, meanings and ways of doing things (Carlile, 2002). For example, professionals from different disciplines and organisations frame problems differently and assess solutions from their own distinct perspectives (Ingerslev, 2016). It is also assumed that in the same cases, professionals from different functional domains have to intensively learn from each other in order to be able to collaborate or co-create (Ingerslev, 2016; Parjanen, 2012a; Schmickl & Kieser, 2008). These differences across boundaries can lead to separation, fragmentation and disconnection, but also to innovation, learning and cross-fertilisation between practices if handled properly (e.g. Carlile, 2004; Kimble et al., 2010; Kislov et al., 2012; Leonard, 1995; Parjanen, 2012a). The purpose of this study is to define the critical elements of collaboration between primary care and the sports sector to better answer the challenge of physical inactivity. The study also sheds light on how this collaboration could be further developed in the relevant organisations to facilitate innovation, learning and cross-fertilisation between primary care and the sports sector.

The role of public health care in physical activity promotion

Physical activity is an essential way of improving people's physical and mental health. It lowers the risks of many diseases and profits society by increasing social interaction and community engagement (Cavill et al., 2006). Physical activity is identified as the most important health-related behaviour that requires change (Leijon et al., 2010). People's participation in physical activity is not only affected by personal issues, but also the built, natural and social environments in which people live. Part of the social environment is public health care.

Public health care has an opportunity to play an important role in encouraging and supporting people to increase their physical activity to a sufficient level in order to achieve good health (Casey, 2007; Puig Ribera et al., 2005; Vuori, 2013). Public health care takes a central position for working towards an increase in physical activity, because many individuals access health care and many also trust it (Professional Association for Physical Activity, 2010; Tulloch et al., 2006; Vuori, 2013). Public health care refers to publicly owned and managed structure for delivering public health operations. However, among countries with public funding of healthcare there are many different approaches to the funding and provision of medical services (Rechel et al., 2018). In Finland, public health care forms the basis of the social welfare and health care system. The system is founded on government-subsidized municipal social welfare and health care services. Health services are divided into primary health care and specialised medical care. Primary health care refers to the municipally arranged services and they include monitoring of the health of the population, promoting wellbeing and health and prevention, diagnosis and treatment of diseases. Specialised medical care refers to secondary and tertiary health care, provided by experts on medical specialties (Ministry of Social Affairs and Health, 2013).

Physical activity promotion through primary care has great potential, but large-scale implementation still faces many challenges (McElwaine et al., 2014; Schmid et al., 2009;

Vuori, 2013). Primary health care currently provides mainly individual services based on acute clinical needs. Treatment goes before prevention and health promotion, and there is likely to be a reluctance among some primary health care professionals to accept health promotion tasks as their responsibilities (Johansson et al., 2010; Vuori, 2013). In addition, primary health care is considered a demanding environment, especially for nurses, who need to have an understanding of many health issues and the ability to take part in health promotion activities (Maijala et al., 2015; Meyer & O'Brien-Pallas, 2010). According to the study by Johansson et al. (2010), many health care professionals state that their knowledge is not used to the advantage of health promotion. In addition, the available evidence for the cost-effectiveness of physical activity interventions is dispersed and varies substantially, and there is uncertainty about the long-term sustainability of behavioural changes (Abu-Omar et al., 2017; Cobiac et al., 2009; Vuori, 2013).

However, there are many opportunities to develop the services already in use in a primary care. There are studies which show that physical activity on prescription is an effective complement or substitute for medication, but it is not used to its full potential (Kallings et al., 2008; Leijon et al., 2009; Persson et al., 2013). The objective of physical activity referral schemes is to identify inactive adults in the primary care setting. The primary care professional refers the patient to a third party service provider. After that, the service provider takes responsibility for prescribing and monitoring a physical activity programme according to the needs of the client. In a primary care setting the physical activity interventions are led by a variety of providers alone or in collaboration, including physicians, nurses and other allied health professionals such as health educators, exercise development officers, physiotherapists, exercise specialists and nutritionists (Tulloch et al., 2006).

According to the literature (Pavey et al., 2012; Williams et al., 2007), physical activity referral schemes have a minor effect on increasing physical activity. There is also evidence that prescribing physical activity is a cost-effective method to use in primary care (Eriksson et al., 2010; Rome et al., 2009). However, there are also doubts is this small benefit an efficient use of resources (Williams et al., 2007). The findings of Rowley et al. (2019) suggest the need to consider physical activity referral schemes and their implementation more critically. For example, the schemes should probably not be targeted at those who are most inactive, because physical activity changes were not sufficient to change the activity category for members of this group.

Physical activity counselling has the potential to produce increases in activity levels (e.g. Altenburg et al., 2015; Olanrewaju et al., 2016; Wu et al., 2019). However, the evidence about which methods or approaches of physical activity promotion work best in the long term is still limited (Lamming et al., 2017; Pavey et al., 2011; Tulloch et al., 2006). Interventions vary widely in terms of what is done within and outside of the health care system, and whether physical activity is promoted individually or in groups (Lion et al., 2019). Although there are studies that show that physical activity referral schemes result in a small improvement in the number of people who increase their levels of physical activity (Campbell et al., 2015; Pavey et al., 2011; Williams et al., 2007) there is still uncertainty about the effectiveness of physical activity referral schemes (Lion et al., 2019; Pavey et al., 2011). This may be partly linked to large variations in different physical activity referral schemes (Lion et al., 2019; Morgan et al., 2016).

Tulloch et al. (2006) endorse an interdisciplinary model in which primary care physicians use their credibility and existing relationship with their patients to suggest increasing physical activity in their daily lives and offer referrals to specialised professionals. According to Persson et al. (2013), nurses are better placed to refer patients to physical activity because physicians

do not have adequate training in non-pharmacological methods, and they do not regard the written referral as a prioritised task and rather refer to other professionals in the health care system to offer referrals. Physician-only interventions usually take place in primary care facilities, while combined-provider and allied health professional-only interventions may occur in various locations. The focus is on promoting physical activity among sedentary patients or patients who could benefit from physical activity, or the target group could be more specific, such as patients with risk factors for certain diseases, like cardiovascular disease or diabetes (Leenaars et al., 2015; Tulloch et al., 2006).

Physical activity counselling and referral could be considered cross-sectoral collaboration (Ingerslev, 2016; Parjanen & Hyypiä, 2018). The collaboration between health care and the sports sector is expected to have the potential to bring about changes in the health of individuals and the population. This does not mean that cross-sectoral collaboration is an easy way of working. In practice, it may mean working in a new area with new working practices and with new people with different backgrounds, knowledge, skills and viewpoints (Casey et al., 2009; Leenaars et al., 2015; Parjanen, 2012a). The literature review of Leenaars et al. (2015) shows that collaboration between primary care and the sports sector is still unexplored. There is a lack of research on partnership-related processes in the collaboration between primary care and the sports sector, and there is a need for more studies that focus on this issue.

Physical activity promotion in the Finnish health care system

In Finland, the prevalence of many diseases that have been common in the past have decreased, but new kinds of health problems have taken their place. For example, lifestyle-related diseases are increasing, and the percentage of overweight adolescents has tripled in the last 30 years. Only one in three gets enough exercise. (Ministry of Social Affairs and Health, 2013.)

In Finland, local authorities are responsible for organising the provision of social welfare and health care services. They can organise the provision of services independently or form joint municipal authorities with each other. In addition, local authorities can outsource the provision of services to other local authorities, a non-governmental organisation or a private service provider. Local authorities can decide on the scale, scope and model of municipal service provision within the limits of legislation. This is why the services available may vary from one municipality to another. (Ministry of Social Affairs and Health, 2013.)

Social welfare and health care services are mostly financed by municipal tax revenue, meaning that the direction of municipal policy rests with the local council. It defines what kind of strategic priority health promotion is for the municipality and incorporates the related objectives in its action plan and budget. The management of health promotion is in essence an exercise of managing processes that cut across administrative boundaries: the challenge is to draw together the knowledge and skills of different branches and to match and coordinate decentralised responsibilities and functions across different administrative sectors. (Quality Recommendation for Health Promotion, 2009.)

According to the Health Care Act, municipalities and joint municipal authorities for hospital districts must develop new practices and consolidate existing practices related to the promotion of health. All age groups must be offered counselling services that promote their well-being, health and functional capacity. Health counselling and checks are a functional entity together with the other services provided by a municipality. Physical activity counselling is a main tool that is used to encourage physically inactive people to increase the levels of physical activity in their daily lives. Health care professionals carry out physical activity counselling.

Effective physical activity counselling is multidisciplinary and cross-administrative. In particular, cooperation between municipal social and health care services, early childhood education and care, and school and sports services is needed. (On the Move, 2013.)

Methods

The research questions for this case study are as follows: What are the critical elements of collaboration between primary care and the sports sector to better answer the challenges of physical inactivity? How could this collaboration be further developed in the relevant organisations? Studying physical activity promotion in the context of primary health care and the sports sector is multidisciplinary and complicated, comprising various professionals and clients in different environments (Ingerslev, 2016; Parjanen & Hyypiä, 2018). The breadth of the case study, its collaborative approach, its acknowledgment of complex contexts, the use of several research methods, its focus on process and outcome and its adaptable approach reflect the benefits of a case study strategy (Yin, 2009).

The case organisation is a public organisation that produces and provides social and healthcare services in five municipalities. It uses the physical activity counselling system. The idea behind this system is that physicians or nurses can recommend physical activity counselling and write a physical activity referral for a patient who they believe would benefit from physical activity. The counselling process provides health screening and advice as well as access to exercise sessions led by qualified physical activity counsellors. Depending on the municipality, the physical activity counsellors work in the municipality's sports department or as project workers in a physical activity-counselling project organised by the regional sports association.

This study uses mixed methods to learn from different perspectives and achieve multiple participation (e.g. Schoonenboom & Johnson, 2017). The purpose of the quantitative data is to offer an overall picture of the research problem, and to help in the formulation of the qualitative data collection. The data was collected from health care professionals, physical activity counsellors and potential clients of physical activity counselling. To answer the research questions about the critical elements of collaboration and the development ideas of physical activity promotion, a questionnaire to health care professionals was conducted. The questionnaire was chosen as a research method because it enables a significant amount of data to be gathered at relatively little cost. It was also considered essential that all employees had the opportunity to share their views. The questionnaire was sent to 127 health care professionals by email, and it consisted of statements concerning physical activity counselling and referral, physical activity counselling in client work, and development ideas related to physical activity counselling and referral. The questionnaire consisted of statements, using a Likert-type rating scale from 1 to 5 (strongly disagree, disagree, neutral, agree and strongly agree). The questionnaire also included questions that allowed for free-form responses, such as what obstacles exist to using referrals, or how can physical activity counselling further developed in their organisation. The researcher, a representative of the organisation and an expert in health promotion devised the questions. Respondents included physicians, nurses, public health nurses and physiotherapists. The response rate was 30.7%.

In addition, two facilitated workshops were organised for nurses and physiotherapists on the above-mentioned themes. These workshops were observed by the researcher. The role of the researcher in the workshops and the meaning of the observation were explained to the participants in the beginning of the workshops. The researcher acted mostly as an observer without participating. In some cases, she assisted the facilitator or took part in the discussions to shorten the social distance between the participants

Table 1. Summary of focus group interviews.

Focus groups	Amount	Time	Content of the interview
Physical activity counsellors	8	50 min	**Key words:** Professional life, things to improve, best memories, difficulties/challenges, strategies to deal with unmotivated clients, perceptions of people who are not physically active **Written question:** What are the most important things or skills that a counsellor should have in promoting physical activity?
Potential clients A	6	55 min	**Key words:** perceptions of physical activity, experiences related to sports or physical activity, barriers of physical activity, motivating factors of physical activity, benefits of physical activity, expectations about health professionals/physical activity counsellors providing health promotion, experiences about health/physical activity professionals
Potential clients B	7	55 min	**Written question:** What elements define a successful encounter with a physical activity professional?

and the observer. When planning the workshops, it was considered essential to establish a trustworthy atmosphere, which helps participants to overcome their reluctance to take a part in the workshop (e.g. Parjanen, 2012b). The purpose of the workshops was to clarify the themes of the questionnaire and offer the opportunity to develop these issues further in a collective manner. As a result, the themes of the questionnaire constituted the protocol of the observation.

To study the opinions and experiences of and ideas about physical activity counsellors and potential clients of physical activity counselling, three focus group interviews were conducted. Focus group interviews were considered more appropriate than individual interviews because the researcher lacks information about the subjects and the intention was to find out about the respondents' understanding. In focus group interviews, people are encouraged to talk to one another. They ask questions, exchange anecdotes and comment on each other's experiences and points of view. In this way, it is possible to explore which issues are important and which are ignored, thus revealing the priorities of the interviewees (Donaldson, 2000; Krueger, 1998; Rabiee, 2004). The focus group interviews have challenges. If people do not feel comfortable sharing their ideas, opinions or experiences, the focus group interview will not generate any useful insights. To avoid this the themes and the meaning of the focus group interview were explained to the participants beforehand and they were also given time to think do they want to participate. Another challenge might be that in the group discussion, quiet participants have not chance to speak because of outspoken participants. To avoid this problem, two interviewers ensured that all the participants have possibility to speak and that all points of view are heard. At the end of the interview, the interviewees had an opportunity to respond in writing to a question about the main topic of the interview. In this study, one focus group interview was conducted with physical activity counsellors (n=8) and two interviews with potential clients (n=13) (Table 1). The interviews included open-ended questions related to the theme. The conversations were tape-recoded and subsequently transcribed.

A thematic analysis was used to analyse the written data. The analysis followed the framework documented by Braun and Clarke (2006): familiarising with data, transcription, generating codes, searching for themes, reviewing themes, defining and naming themes and producing the report. Thought the framework is presented as a linear, the analysis was an iterative and reflective process involving a constant moving back and forward between phases (Nowell et al., 2017). The themes were generated inductively, allowing the data to determine the themes. Firstly, the data was analysed so that the factors enhancing or hindering collaboration were revealed. In the second phase, the data was analysed in more detail so that sub-themes were constructed. To answer the first research

question, the following sub-themes were generated: organisational support, knowledge sharing, skills, evaluation, stakeholders, and best practices. To answer the second research question, one sub-theme was found: development ideas. Then, the results of the analysis were compared with the research literature. In addition, during the analysis the researcher considered whether the data sets collected by different methods differed or supported each other.

Results

The present state in the organisation

Based on the results of the questionnaire, the health care professionals considered physical activity counselling to be part of health promotion. None disagreed with the statement. As seen in Table 2, during client visits, 66 per cent of the respondents ask almost every client about their physical activity habits. Some 64 per cent of the respondents refer a client to physical activity counselling at least once a month. Over 70 per cent of the respondents ask about physical activity counselling if the client has been given a referral. The respondents considered, for example, that "physical activity referral supports further rehabilitation with physiotherapy. It is a good opportunity to improve one's ability to function and find regular everyday activities and sport", or "it is a good tool to get people to move and think about changes in their way of living". However, according to the respondents, the opportunities for physical activity counselling are not fully utilised. For example, respondents considered that more clients could benefit from physical activity counselling.

One quarter of the respondents did not take a stand on the question of whether they cooperate with a physical activity counsellor (Table 3). In the open answer, a respondent explained, "It would be nice to hear about what kinds of wishes and experiences the physical activity counsellors have about current practices and in general it would be good to meet them, for example, a couple times a year". There was also a clear wish in the workshops that collaboration between physical activity counselling and physiotherapy should be closer. One respondent said that "cooperation between physical activity counsellors and health care professionals is one of the most important issues" in a successful physical activity referral process. The abovementioned

Table 2. The statement: How many clients do you ask physical activity habits? (%).

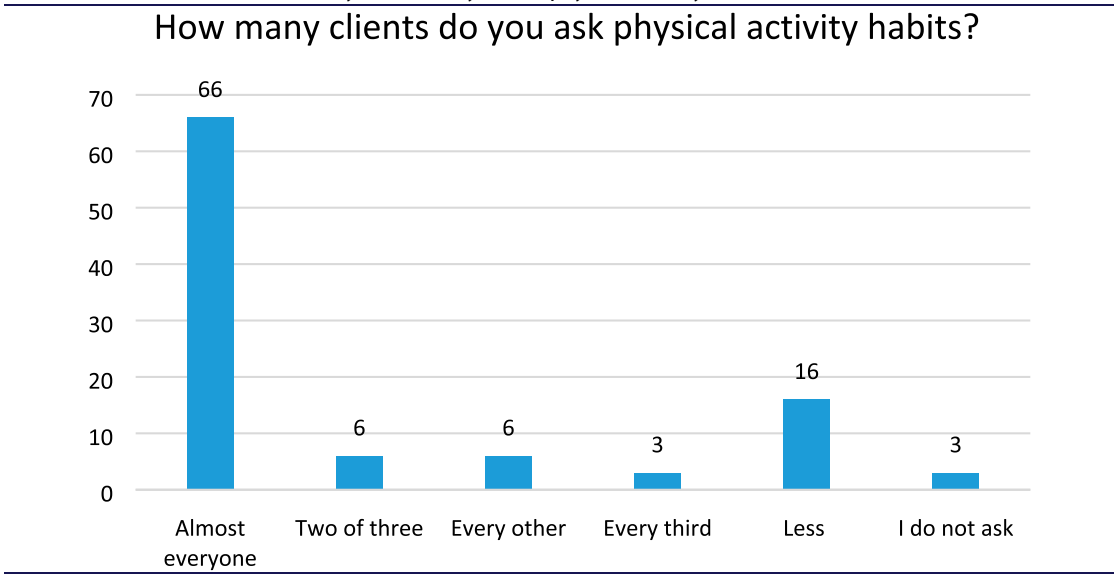
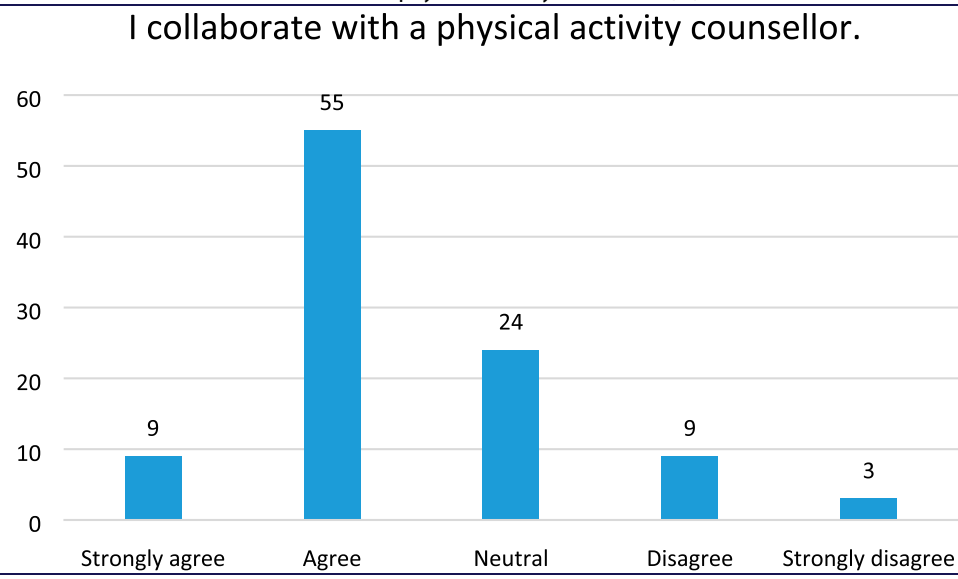


Table 3. The statement: I collaborate with the physical activity counsellor (%).

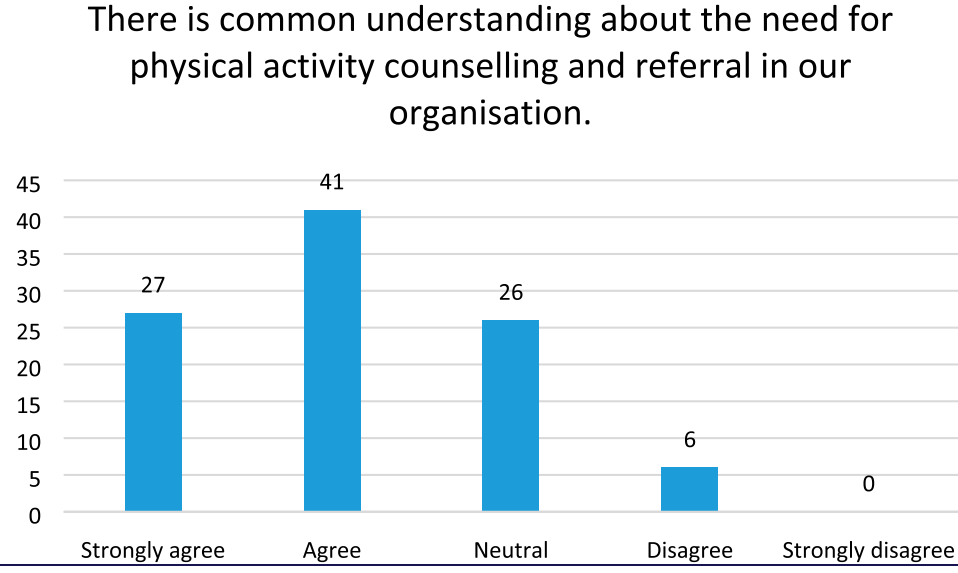


issues imply that there are boundaries between different professionals in the organisation.

The need for managerial support

Most of the respondents considered that there is a common objective in the organisation regarding the need for physical activity counselling and referral (Table 4). However, many health care professionals considered that they would like to have more discussions about the role of physical activity counselling and referral in health promotion. This was considered the responsibility of the management. One

Table 4. The statement: There is common understanding about the need for physical activity counselling in our organisation (%).



respondent explained that the organisation "should give a clear presentation about why physical activity counselling is important in the organisation and how it should be given". Based on the focus group interview of physical activity counsellors, it was difficult to define what kind of managerial support they would need. In fact, they ignored the question about managerial support. This could be explained by their role of being between two organisations and confusion about managerial roles in this situation. They very strongly stressed the role of collegial support in their work. They appreciated the opportunity to ask for help and discuss the problems in their work with other physical activity counsellors.

Knowledge sharing between professionals

Many respondents found that there are many things that should be developed between the social and health care sector and the sports department (Table 5). Collaboration depends on the ability to coordinate different kinds of knowledge within the organisation (for example between different units) and between organisations. There is still a need, for example, to "inform personnel more loudly and visibly, and through various channels, about the fact that physical activity counsellors are based in health centres" and a need to organise "common meetings with various actors".

Actually, many considered that their responsibilities and duties were unclear (Table 6), causing problems, for example, in knowledge sharing. There was confusion about who is responsible for communication regarding issues related to physical activity promotion and with whom one should share knowledge. One respondent stressed that the "administration of health promotion should be in health care and actually in one area of responsibility. The area of responsibility should be one that in practice also takes care of the patient, not one that theoretically only stands by". According to this respondent, it is important that responsibility for physical activity promotion is organised so that there is "a clear connection between clients and practice, because this would allow activities to have more of an impact".

The respondents felt that they have enough knowledge about physical activity counselling and that the training has been good. Only a

Table 5. The statement: Physical activity referral practices should be developed more (%).

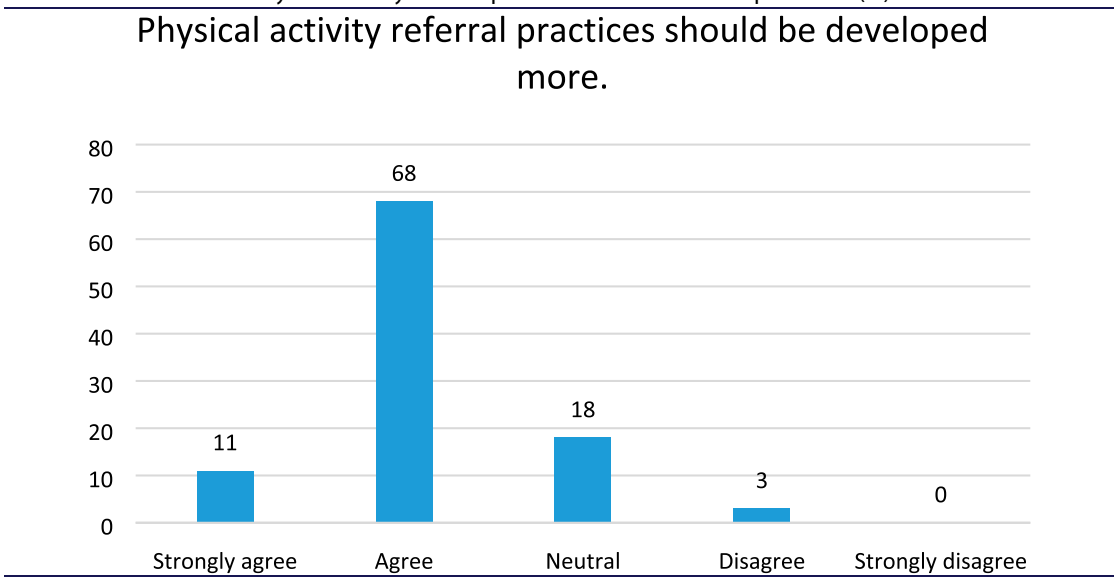
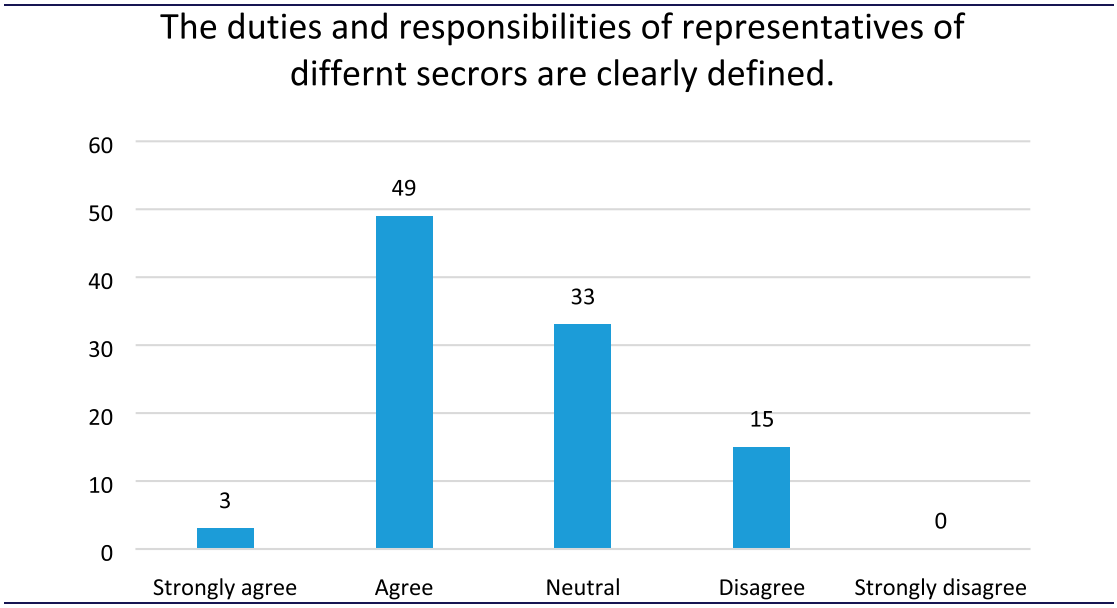


Table 6. The statement: The duties and responsibilities of representatives of different sectors are clearly defined (%).



few considered that there should be more training. In many of the open answers, there were references to how organising the training. There should be basic information about health promotion and physical activity counselling, but also opportunities to meet other professionals and discuss with them about their points of view on physical activity counselling. There was an obvious need to share explicit knowledge about the effects of physical activity counselling, but also a need to share tacit knowledge, such as experiences about challenging clients. The respondents would have liked to have had different kinds of example cases and discussions based on these examples about who benefits from physical activity counselling and how different clients benefit from counselling.

The evaluation of effectiveness of physical activity counselling

It was widely considered that the evaluation of effectiveness of physical activity counselling was important. In general, there was a need to move "from quantitative to more qualitative measurement" as one of the respondent explained in her free-form response. Some of the respondents thought that it is essential to stress the benefits that the client will experience. Some of the health care professionals questioned whether the clients have actually benefited from physical activity counselling. Thirty-five per cent of the respondents did not take a stand on the statement "The clients have benefited from physical activity counselling". Two respondents disagreed with the statement. This could imply that feedback on the physical activity counselling process is missing. In fact, the physical activity counsellors stressed that it is important that health care professionals ask the client about the physical activity counselling process or read comments from the counsellor in the data system.

Attitudes, skills and personal experiences of the health care professionals

According to the physical activity counsellors, it is important that health care professionals have

the skills to select those clients who have the opportunities to benefit from a physical activity counselling process. Physical activity counselling is not the right thing for every client and it should not be considered as "the last chance" after everything else has failed. For example, many physical activity counsellors highlighted that it is essential that health care professionals can see the difference between clients that need physiotherapy and who benefit from physical activity counselling. Physical activity counsellors also valued good background information about the client. Good basic information was highlighted in cases where the client has mental health problems, for example. Especially in the case of mental illness, physical activity counsellors stressed that they are not experts in mental illness and it would be a good idea if in demanding cases there was also somebody from the referring organisation. One reason for this is that the physical activity counsellors considered medical concepts difficult, as one counsellor says that physicians and physical activity counsellors "do not speak the same language".

Health care professionals felt that it is difficult to bring up physical activity with the client because clients are usually unwilling to talk about physical activity. Health care professionals also highlighted that sometimes personal attitudes avoided bringing up physical activity with the client. In that way there is a risk that physical activity promotion focuses on selected clients. According to one free-form response, "the most important thing is that the professionals themselves believe in the possibilities of physical activity". It is also important that health care professionals have knowledge about physical activity counselling itself, including who the physical activity counsellors are and what happens during the counselling process. According to the physical activity counsellors, health care professionals lack knowledge of the expertise that physical activity counsellors hold. Also, the lack of personal experience about physical activity counselling may hinder the use of counselling referrals in general. One respondent had visited a physical activity counsellor and explained that "it is now easier to encourage clients to accept a physical activity referral when I know myself how the counselling works".

One of the most commonly-mentioned barriers according to the health care professionals was reluctant clients. The health care professionals though that many clients do not want to talk about physical activity because they had bad experiences of physical education at school, or they are afraid of attending an exercise group. The health care professionals worried that that these real reasons never come up in the conversation. However, it is also essential to understand how health care professionals tell their clients about physical activity counselling. One physical activity counsellor explained that there is a physician who is very enthusiastic about physical activity counselling, and when calling his/her clients, the physical activity counsellor can hear from the client's voice that they feel "like they have won the lottery by having the opportunity to have personal counselling". At the other extreme, there are clients who have no idea what they have been referred to. There is also the possibility that if the client experiences that they are forced into the counselling, they will never contact the physical activity counsellor.

Knowledge and social skills in counselling

According to the physical activity counsellors, it is good to give some time to clients before the first appointment with the physical activity counsellor. During that time, the client has an opportunity to think about his/her own physical activity. The physical activity counsellors stressed that usually those clients who have had time to think about their own need to increase physical activity are more motivated. During that time, the physical activity counsellor may contact the client by phone and discuss issues that occupy the client's mind.

The physical activity counsellors described the first phase with the client as "breaking the ice". The more negative an attitude the client has, the more demanding this phase is. Most of the counselling process is about mapping the client's situation in life, and social skills are especially highlighted. It is not enough to know the benefits of physical activity; the physical activity counsellor should also be a good listener and have skills to pull "the right strings". Actually, the physical activity counsellors explained that they should have the knowledge of "physicians, therapists and life coaches". According to the physical activity counsellors, they should have knowledge of the benefits of physical activity, the basic facts about nutrition, and knowledge about sports opportunities in the region. They also listed many characteristics that they need in their work, such as openness, friendliness, enthusiasm, patience, reliability, ability to motivate and a positive attitude. They also highlighted skills such as using personal life experience, understanding the client, encountering the clients as they are, and giving them time. In fact, these are the same things that the potential clients considered essential for a successful encounter with the counsellor.

The potential clients stressed that "professional knowledge is not enough". Instead, "attitude", "personality" and "genuine interest" are essential. The clients want to feel that they are heard. The clients were also afraid that the counsellor may be "too athletic" or if the counsellor "does not understand the clients' restrictions regarding physical activity". Physical activity was considered a very personal thing, and using the wrong words might cause resentment. The clients also stressed that it is important to have understandable short-term targets. It is good to not have too many goals at the same time, and the wishes of the client have to be respected. These are the starting points. At its best, the client finds their own solutions to the problem. As one counsellor said, "sometimes it means discussing everything else other than physical activity".

Development ideas about physical activity counselling

Because of the large number of inactive people, physical activity referrals will need to become more of a specialist resource for primary health care. To achieve this, the criteria for physical activity referral should be more clearly defined among various professionals. This need was considered one of the most urgent issues to be agreed on.

In addition, there was a clear need to evaluate the physical activity promotion process continuously, thus highlighting issues that need development. The evaluation provides an insight into the process of implementation and gives information on what went on during the process. It can help to identify factors contributing to success or failure. It is essential to collect evaluation data from different stakeholders, such as health care professionals, physical activity counsellors, managers, clients, sports clubs, administrators and policy-makers.

According to the data, dialogue between health professionals and physical activity counsellors would enhance successful health promotion. For example, the idea of physical activity counsellors attending meetings with health care professionals was considered fruitful. For further development, it would be beneficial to define those issues where different kinds of expertise are needed. For example, one health care professional felt that in the development of referrals the expertise of physicians is needed. In addition, there was a need to develop new tools for physical activity counselling. Some of these tools could also be virtual, utilising technology.

A successful physical activity counselling process is also dependent on locally provided physical activity opportunities. None of the professionals wanted a situation where "the client feels they were being sold down the river due to a lack of physical activity opportunities". There is a need for easily accessible groups for beginners and senior citizens. In the free-form

responses, there were suggestions about opening and enlarging the network that develops health promotion and physical activity counselling. In addition to local sports clubs, the network could also include representatives from the municipal council, childhood education, schools, third sector organisations and the local media. The role of the local media was considered essential as a link to the wider audience in the region. Sixty-eight per cent of the health care professionals thought that clients do not know enough about physical activity counselling. This was confirmed in the client focus group interviews. The interviewees had not heard about physical activity counselling. Neither did they know about physical activity counsellors. Figure 1 presents the critical elements for cross-sectoral collaboration in the context of a physical activity referral process.

Discussion

Physical activity counselling was seen as a collaboration between many experts from different fields of operation. In practice, it is collaboration between health care professionals and physical activity counsellors. The results of this study are consistent with the study of Persson et al. (2013), in that the competencies of different professions are utilised to achieve optimum teamwork in physical activity promotion. In addition, this study highlights the social proximity between different professionals that cross functional and professional boundaries (Ingerslev, 2016; Parjanen, 2012a; Schmickl & Kieser, 2008). Professionals that know each other and trust each other's competencies collaborate, because it is possible that intra-organisational boundaries across departments are sometimes more tightly defined than relationships across organisations.

According to this study, the health care professionals understand that the promotion of physical activity is an essential part of their responsibilities. This differs from the existing literature about the reluctance to accept health promotion tasks as the responsibilities of primary health care (Johansson et al., 2010; Vuori, 2013). However, the questionnaire's response rate was only 30.7 per cent, so it is possible that those who did not answer the questionnaire would have different opinions related to the theme or their willingness to use physical activity counselling and referral might vary.

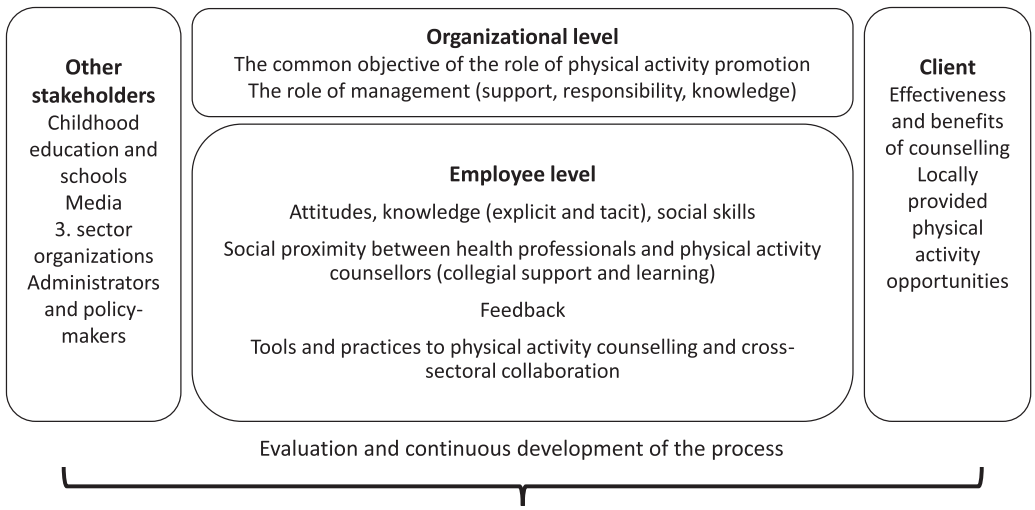


Figure 1. The critical elements for cross-sectoral collaboration in the context of a physical activity counselling and referral process.

According to the results, health promotion counsellors need different kinds of knowledge and skills. For example, strong communication skills and unbiased interaction with a variety of clients in health promotion has already been stated in the existing literature (Goodman et al., 2011; Maijala et al., 2015). In addition, this study supports the findings of Casey et al. (2016), which suggest that in order to promote cross-sectoral collaboration, there is a need to develop a shared language between professionals and across different sectors. For example, the physical activity counsellors felt that medical concepts were difficult to grasp, and that this hindered collaboration between physicians and counsellors (see also Foley et al., 2000; Wiles et al., 2008). This also raises a question about how effectively new professional requirements meet the content of current sports education and learning programmes (see also Campos-Izquierdo et al., 2016).

Most of the health care professionals and physical activity counsellors considered that physical activity counselling requires more development. For example, the possibilities of physical activity counselling are not fully utilised – something that is also supported by earlier studies (Kallings et al., 2008; Leijon et al., 2009; Persson et al., 2013). Thus, an essential question is how to evaluate the effectiveness of the process at an individual level but also at the organisational level. It was felt that the physical activity counselling process needs continuous development efforts to better respond to changes in the society. It is an iterative process of incorporating both modest, incremental and radical improvements in processes and services. The evaluation is important for finding out unworkable practices and validating the necessary resources for policy-makers.

There was also a need for new tools, including virtual ones, to make counselling more effective. This would probably mean that physical activity professionals would also need more training on how to use IT and social media in physical activity counselling. In addition, new practices for making cross-sectoral collaboration more flexible were also needed. In the previous literature, it is suggested that social proximity can be developed through interprofessional training programmes – to bring professionals from different fields together and increase the legitimacy of multiple forms of data, information, tools and methods (Casey et al., 2016). This study also implies that managers should create opportunities for their employees to share experimental knowledge, such as client contact know-how, and use the ideas of the employees in the development of training.

In the existing literature, the availability of sports infrastructure influences patterns of sports activity significantly (Hallmann et al., 2012; Wicker et al., 2009). More information about sports opportunities in the region were needed, especially by the health care professionals, and sports clubs and third sector organisations were considered essential stakeholder groups. The results of this study highlight that increasing physical activity is a societal challenge rather than just an organisational one, meaning that larger networks of different actors, like administrators, policy-makers, schools and the media, are needed.

The common objective of the role of physical activity promotion builds the foundations for a successful referral process. Managers have an essential role to play in this. The role of the management in successful physical activity promotion is to maintain a conversation about the benefits of physical activity promotion and remind employees about the importance of physical activity promotion. It is their responsibility to explain why physical activity promotion is important in the organisation, and to maintain the dialogue about promotion. It is also essential that the managers themselves are the ones who understand physical activity promotion and are essential for driving forward and assisting the change implementation of the new practices. This study identifies managerial actions, such as communication about the essence of physical activity promotion, the

creation of clear roles and responsibilities, the facilitation of collaboration between professionals, and employee-driven training to support successful knowledge creation.

Conclusion

Working cross-sectorally represents a key opportunity for creating a functioning physical activity counselling process. This means that finding solutions will require more open, honest and constructive ways of working together. It requires trust, communication, and a shared understanding of purpose and outcomes. According to this case study, a common organisational objective in physical activity promotion creates the foundation for a successful physical activity counselling and referral process. The common objective should be supported by the management. It is essential that there are discussions about the role of physical activity counselling between management and employees. In the data, the social distance between health care professionals and physical activity counsellors became very clear. Health care professionals hesitate to refer clients to physical activity counsellors because they do not know any physical activity counsellors personally or they are not familiar with the content or methods of the physical activity counselling process. The study suggests that the realisation of a physical activity counselling process requires greater development at multiple levels, and that interaction between multiple stakeholders is closer. It takes place in networks and is collective by nature. The physical activity counselling process should not be considered as a static path; the process needs to be continuously evaluated based on objectives formulated by the stakeholders.

Multiple research opportunities may be explored in the future. The physical activity counsellors highlighted the need to study the physical activity counselling process and its effects on changes in behaviour more carefully. More comprehensive descriptions of the process would underline the main factors of an effective process from a client's and a counsellor's perspectives. It is also essential to gain feedback on clients' experiences to help understand the factors influencing adherence and drop-out, and to inform developments to the implementation of the scheme. In addition, it would be interesting to study the possibilities of group counselling and the role of peer support in the context of physical activity promotion. Digitalisation also brings about new opportunities, and there is a growing literature that emphasises the possibilities that mediated channels have for delivering physical activity advice and information. This may bring about new kinds of demands on counselling and counsellors.

Acknowledgements

The author thanks the co-workers in Motion! project who gave valuable ideas and comments for planning and carrying out the questionnaire, focus group interviews and workshops. The author is grateful for the health care professionals, physical activity counsellors and potential clients of physical activity counselling who answered the questionnaire and participated in focus group interviews and workshops. The Motion! Project was funded by the European Regional Development Fund and the Regional Council of Päijät-Häme. Amer Cultural Foundation supported the manuscript preparation phase.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

The author thanks the co-workers in Motion! project who gave valuable ideas and comments for planning and carrying out the questionnaire, focus group interviews and workshops. The author is grateful for the health care professionals, physical activity counsellors and potential clients of physical activity counselling who answered the questionnaire and participated in focus group interviews and workshops. The Motion! Project was funded by the European

Regional Development Fund and the Regional Council of Päijät-Häme. Amer Cultural Foundation supported the manuscript preparation phase.

References

Abu-Omar, K., Rütten, A., Burlacu, I., Schätzlein, V., Messing, S., & Suhrcke, M. (2017). The cost-effectiveness of physical activity interventions: A systematic review of reviews. *Preventive Medicine Reports*, *8*, 72–78. https://doi.org/10.1016/j.pmedr.2017.08.006

Allender, S., Foster, C., Scarborough, P., & Rayner, M. (2007). The burden of physical activity-related ill health in the UK. *Journal of Epidemiology and Community Health*, *61*, 344–348. https://doi.org/10.1136/jech.2006.050807

Altenburg, W. A., ten Hacken, N. H. T., Bossenbroek, L., Kerstjens, H. A. M., de Greef, M. H. G., & Wempe, J. B. (2015). Short- and long-term effects of a physical activity counselling programme in COPD: A randomized controlled trial. *Respiratory Medicine*, *109* (1), 112–121. https://doi.org/14.10.020

Booth, F. W., Roberts, C. K., & Laye, M. J. (2012). Lack of exercise is a major cause of chronic diseases. *Comprehensive Physiology*, *2*(2), 1143–1211. https://doi.org/10.1002/cphy.c110025

Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, *3*, 77–101. https://doi.org/10.1191/1478088706qp063oa

Campbell, F., Holmes, M., Everson-Hock, E., Davis, S., Buckley Woods, H., Anokye, N., Tappenden, P., & Kaltenthaler, E. (2015). A systematic review and economic evaluation of exercise referral schemes in primary care: A short report. *Health Technology Assessment (Winchester, England)*, *19*(60), 1–110. https://doi.org/10.3310/hta19600

Campos-Izquierdo, A., González-Rivera, M., & Taks, M. (2016). Multi-functionality and occupations of sport and physical activity professionals in Spain. *European Sport Management Quarterly*, *16*(1), 106–126. https://doi.org/10.1080/16184742.2015.1108990

Carlile, P. R. (2002). A pragmatic view of knowledge and boundaries: Boundary objects in new product development. *Organization Science*, *13*(4), 442–455. https://doi.org/10.1287/orsc.13.4.442.2953

Carlile, R. P. (2004). Transferring, translating, and transforming: An integrative framework for managing knowledge across boundaries. *Organization Science*, *15*(5), 555–568. https://doi.org/10.1287/orsc.1040.0094

Casey, D. (2007). Nurses' perception, understanding and experiences of health promotion. *Journal of Clinical Nursing*, *16*, 1039–1049. https://doi.org/10.1111/j.1365-2702.2007.01640.x

Casey, C., Li, J., & Berry, M. (2016). Interorganizational collaboration in public health data sharing. *Journal of Health Organization and Management*, *30*(6), 855–871. https://doi.org/10.1108/JHOM-05-2015-0082

Casey, M., Payne, W., & Eime, R. (2009). Partnership and capacity-building strategies in community sports and recreation programs. *Managing Leisure*, *14*, 167–176. https://doi.org/10.1080/13606710902944938

Cavill, N., Kahlmeier, S., & Racioppi, F. (Eds.) (2006). *Physical activity and health in Europe: Evidence for action*. WHO Regional Office for Europe. http://www.euro.who.int/__data/assets/pdf_file/0011/87545/E89490.pdf

Cobiac, L. J., Vos, T., & Barendregt, J. J. (2009). Cost-effectiveness of interventions to promote physical activity: A modelling study. *PLOS Medicine*, *6*(7), e1000110. https://doi.org/10.1371/journal.pmed.1000110

Donaldson, L. J. (2000). Sport and exercise: The public health challenge. *British Journal of Sports Medicine*, *34*, 409–415. https://doi.org/10.1136/bjsm.34.6.409

Eriksson, M. K., Hagberg, L., Lindholm, L., Malmgren-Olsson, E. B., Osterlind, J., & Eliasson, M. (2010). Quality of life and cost-effectiveness of a 3-year trial of lifestyle intervention in primary health care. *Archives of Internal Medicine*, *170*, 1470–1479. https://doi.org/10.1001/archinternmed.2010.301

Foley, M., Frew, M., McPherson, G., & Reid, G. (2000). Healthy public policy: A policy paradox within local government. *Managing Leisure*, *5*, 77–90. https://doi.org/10.1080/13606710050022539

Garrett, N., Brasure, M., Schmitz, K. H., & Huber, M. R. (2004). Physical inactivity: Direct cost to a health plan. *American Journal of Preventive Medicine*, *27*, 304–309. https://doi.org/10.1016/j.amepre.2004.07.014

Goodman, C., Davies, S. L., Dinan, S., See, T. S., & Iliffe, S. (2011). Activity promotion for community-dwelling older people: A survey of the contribution of primary care nurses. *British Journal of Community Nursing*, *16*, 12–17. https://doi.org/10.12968/bjcn.2011.16.1.12

Hagberg, L. (2007). *Cost-effectiveness of the promotion of physical activity in health care*. Umeå University Medical Dissertations, New Series No 1085, Umeå University, Umeå.

Hallmann, K., Wicker, P., Breuer, C., & Schönherr, L. (2012). Understanding the importance of sport infrastructure for participation in different sports – findings from multi-level modeling. *European Sport Management Quarterly*, *12*(5), 525–544. https://doi.org/10.1080/16184742.2012.687756

Husu, P., Paronen, O., Suni, J., & Vasankari, T. (2011). *Suomalaisten fyysinen aktiivisuus ja kunto 2010. Terveyttä edistävän liikunnan nykytila ja muutokset*. Opetus- ja kulttuuriministeriön julkaisuja 2011:15.

Ingerslev, K. (2016). Crossing and creating boundaries in healthcare innovation. *Journal of Health Organization and Management*, *30*(4), 541–557. https://doi.org/10.1108/JHOM-07-2015-0107

Johansson, H., Stenlund, H., Lundström, L., & Weinehall, L. (2010). Reorientation to more health promotion in health services – a study of barriers and responsibilities from the perspective of health professionals. *Journal of Multidisciplinary Healthcare*, *3*, 213–224. https://doi.org/10.2147/JMDH.S14900

Kallings, L. V., Leijon, M., Hellénius, M.-L., & Ståhle, A. (2008). Physical activity on prescription in primary health care: A follow-up of physical activity level and quality of life. *Scandinavian Journal of Medicine & Science in Sports*, *18*, 154–161. https://doi.org/10.1111/j.1600-0838.2007.00678.x

Kimble, C., Grenier, C., & Goglio-Primard, K. (2010). Innovation and knowledge sharing across professional boundaries: Political interplay between boundary objects and brokers. *International Journal of Information Management*, *30*(5), 437–444. https://doi.org/10.1016/j.ijinfomgt.2010.02.00

Kislov, R., Walshe, K., & Harvey, G. (2012). Managing boundaries in primary care service improvement: A developmental approach to communities of practice. *Implementation Science*, *7*, 97. https://doi.org/10.1186/1748-5908-7-97

Krueger, R. (1998). *Analyzing and reporting focus group results (focus group kit)*. Sage Publications.

Lamming, L., Pears, S., Mason, D., Morton, K., Bijker, M., Sutton, S., & Hardeman, W. (2017). What do we know about brief interventions for physical activity that could be delivered in primary care consultations? A systematic review of reviews. *Preventive Medicine*, *99*, 152–163. https://doi.org/10.1016/j.ypmed.2017.02.017

Leenaars, K. E. F., Smit, E., Wagemakers, A., Molleman, G. R. M., & Koelen, M. A. (2015). Facilitators and barriers in the collaboration between the primary care ant the sport sector in order to promote physical activity: A systematic literature review. *Preventive Medicine*, *81*, 460–478. https://doi.org/10.1016/j.ypmed.2015.10.010

Leijon, M. E., Bendtsen, P., Nilsen, P., Festin, K., & Stahle, A. (2009). Does a physical activity referral scheme improve the physical activity among routine primary health care patients? *Scandinavian Journal of Medicine & Science in Sports*, *19*, 627–636. https://doi.org/10.1111/j.1600-0838.2008.00820.x

Leijon, M. E., Stark-Ekman, D., Nilsen, P., Ekberg, K., Walter, L., Stahle, A., & Bendtsen, P. (2010). Is there a demand for physical activity interventions provided by the health care sector? Findings from a population survey. *BMC Public Health*, *10*. https://doi.org/10.1186/1471-2458-10-34

Leonard, D. (1995). *Wellsprings of knowledge. Building and sustaining the sources of innovation*. Harvard Business School Press.

Lion, A., Vuillemin, A., Thornton, J. S., Theisen, D., Stranges, S., & Ward, M. (2019). Physical activity promotion in primary care: A Utopian quest? *Health Promotion International*, *34*(4), 877–886. https://doi.org/10.1093/heapro/day038

Maijala, V., Tossavainen, K., & Turunen, H. (2015). Identifying nurse practitioners' required case management competencies in health promotion practice in municipal public primary health care. A two-stage modified Delphi study. *Journal of Clinical Nursing*, *24*, 2554–2561. https://doi.org/10.1111/jocn.12855

McElwaine, K., Freund, M., Campbell, E. M., Slattery, C., Wye, P. M., Lecathelinais, C., Bartlem, K. M., Gillham, K. E., & Wiggers, J. H. (2014). Clinician assessment, advice and referral for multiple health risk behaviors: Prevalence and predictors of delivery by primary health care nurses and allied health professionals. *Patient Education and Counseling*, *94*, 193–201. https://doi.org/10.1016/j.pec.2013.10.017

Meyer, R. M., & O'Brien-Pallas, L. L. (2010). Nursing services delivery theory: An open system approach. *Journal of Advanced Nursing*, *66*, 2828–2838. https://doi.org/10.1111/j.1365-2648.2010.05449.x

Ministry of Social Affairs and Health. (2013). *Health care in Finland*. Brochures of the Ministry of Social Affairs and Health. Juvenes Print – Finnish University Print Ltd.

Morgan, F., Battersby, A., Weightman, A. L., Searchfield, L., Turley, R., Morgan, H., Jagroo, J., & Ellis, S. (2016). Adherence to exercise referral schemes by participants – what do providers and commissioners need to know? A systematic review of barriers and facilitators. *BMC Public Health*, *16*, 227. https://doi.org/10.1186/s12889-016-2882-7.

Nowell, L. S., Norris, J. M., White, D. E., & Moules, N. J. (2017). Thematic analysis: Striving to meet the

trustworthiness criteria. *International Journal of Qualitative Methods*, *16*, 1–13. https://doi.org/10.1177/1609406917733847

Olanrewaju, O., Kelly, S., Cowan, A., Brayne, C., & Lafortune, L. (2016). Physical activity in community dwelling older people: A systematic review of reviews of interventions and context. *PloS one*, *11* (12), e0168614. https://doi.org/10.1371/journal.pone.0168614

On the Move. (2013). *National strategy for physical activity promoting health and wellbeing 2020*. Publications of the Ministry of Social Affairs and Health. http://www.julkari.fi/bitstream/handle/10024/110661/978-952-00-3417-7_korj.pdf

Parjanen, S. (2012a). *Creating possibilities for collective creativity. Brokerage functions in practice-based innovation*. Acta Universitatis Lappeenrantaensis 474, Yliopistopaino.

Parjanen, S. (2012b). Innovation sessions as sources of new ideas. *International Journal of Innovation and Learning*, *11*(4), 352–368. https://doi.org/10.1504/IJIL.2012.047137

Parjanen, S., & Hyypiä, M. (2018). Innovation platforms as a solution to the proximity paradox. *European Planning Studies*, *26*, 1312–1329. https://doi.org/10.1080/09654313.2018.1476469

Pavey, T. G., Taylor, A. H., Fox, K. R., Hillsdon, M., Anokye, N., Campbell, J. L., Foster, C., Green, C., Moxham, T., Mutrie, N., Searle, J., Trueman, P., & Taylor, R. S. (2011). Effect of exercise referral schemes in primary care on physical activity and improving health outcomes: Systematic review and meta-analysis. *BMJ*, *343*, d6462. https://doi.org/10.1136/bmj.d6462

Pavey, T., Taylor, A., Hillsdon, M., Fox, K., Campbell, J., Foster, C., Moxham, T., Mutrie, N., Searle, J., & Taylor, R. (2012). Levels and predictors of exercise referral scheme uptake and adherence: A systematic review. *Journal of Epidemiology and Community Health*, *66*, 737–744. https://doi.org/10.1136/jech-2011-200354

Persson, G., Brorsson, A., Ekvall Hansson, E., Troein, M., & Strandberg, E. L. (2013). Physical activity on prescription (PAP) from the general practitioner's perspective – a qualitative study. *BMC Family Practice*, *14*, 128. https://doi.org/10.1186/1471-2296-14-128

Professional Associations for Physical Activity. (2010). *Physical activity in the prevention and treatment of disease*. Swedish National Institute of Public Health. http://www.fyss.se/wp-content/uploads/2011/02/fyss_2010_english.pdf

Puig Ribera, A., McKenna, J., & Riddoch, C. (2005). Attitudes and practices of physicians and nurses regarding physical activity promotion in the Catalan primary health-care system. *European Journal of Public Health*, *15*, 569–575. https://doi.org/10.1093/eurpub/cki045

Quality Recommendation for Health Promotion. (2009). Publications of the Ministry of Social Affairs and Health, Helsinki University Print, Helsinki, Finland.

Rabiee, F. (2004). Focus-group interview and data analysis. *Proceedings of the Nutrition Society*, *63*, 655–660. https://doi.org/10.1079/PNS2004399.

Rechel, B., Maresso, A., Sagan, A., Hernández-Quevedo, C., Williams, G., Richardson, E., Jakubowski, E., & Nolte, E. (2018). *Organization and financing of public health services in Europe: Country reports*. European Observatory on Health Systems and Policies, Health Policy Series, No. 49.

Rome, Å, Persson, U., Ekdahl, C., & Gard, G. (2009). Physical activity on prescription (PAP): Costs and consequences of a randomized, controlled trial in primary healthcare. *Scandinavian Journal of Primary Health Care*, *27*, 216–222. https://doi.org/10.3109/02813430903438734

Rowley, N., Steele, J., Wade, M., Copeland, R. J., Mann, S., Liguori, G., Horton, E., & Jimenez, A. (2019). Are exercise referral schemes associated with an increase in physical activity? Observational findings using individual patient data meta-analysis from the National Referral Database. https://doi.org/10.31236/osf.io/ckdwn

Schmickl, C., & Kieser, A. (2008). How much do specialists have to learn from each other when they jointly develop radical product innovations? *Research Policy*, *37*(3), 473–491. https://doi.org/10.1016/j.respol.2007.12.001

Schmid, M., Egli, K., Martin, B. W., & Bauer, G. (2009). Health promotion in primary care: Evaluation of a systematic procedure and stage specific information for physical activity counselling. *Swiss Medical Weekly*, *139*, 665–671. https://doi.org/smw-12680

Schoonenboom, J., & Johnson, R. B. (2017). How to construct a mixed methods research design. Wie man ein mixed methods-Forschungs-design konstruiert. *KZfSS Kölner Zeitschrift für Soziologie und Sozialpsychologie*, *69*(2), 107–131. https://doi.org/10.1007/s11577-017-0454-1

Thorp, A., Owen, N., Neuhaus, M., & Dunstan, D. (2011). Sedentary behaviors and subsequent health outcomes in adults. A systematic review of longitudinal studies, 1996–2011. *American Journal of Preventive Medicine*, *41*(2), 207–215. https://doi.org/10.1016/j.amepre.2011.05.004

Tulloch, H., Fortier, M., & Hogg, W. (2006). Physical activity counseling in primary care: Who has and

who should be counselling? *Patient Education and Counseling*, *64*, 6–20. https://doi.org/10.1016/j.pec.2005.10.010

Vuori, I. (2013). Role of primary health care in physical activity promotion. *Deutsche Zeitschrift für Sportmedizin*, *64*, 176–182. https://doi.org/10.3109/13814788.2012.675504

Wicker, P., Breuer, C., & Pawlowski, T. (2009). Promoting sport for all to age-specific target groups: The impact of sport infrastructure. *European Sport Management Quarterly*, *9*(2), 103–118. https://doi.org/10.1080/16184740802571377

Wiles, R., Demain, S., Robison, J., Kileff, J., Ellis-Hill, C., & McPherson, K. (2008). Exercise on prescription schemes for stroke patients post-discharge from physiotherapy. *Disability and Rehabilitation*, *30*, 1966–1975. https://doi.org/10.1080/09638280701772997

Williams, N. H., Hendry, M., France, B., Lewis, R., & Wilkinson, C. (2007). Effectiveness of exercise-referral schemes to promote physical activity in adults: Systematic review. *British Journal of General Practice*, *57*(545), 979–986. https://doi.org/10.3399/096016407782604866

World Health Organisation (WHO). (2010). *Global recommendations on physical activity for health*. http://apps.who.int/iris/bitstream/10665/44399/1/9789241599979_eng.pdf

Wu, M. L., Tsai, J. C., Yu, K. H., & Chen, J. J. (2019). Effects of physical activity counselling in women with systemic lupus erythematosus: A randomized controlled trial. *International Journal of Nursing Practice*, *25*(5), e12770. https://doi.org/10.1111/ijn.12770

Yin, R. (2009). *Case study research. Design and methods*. Applied social research methods series (3rd ed., Vol. 5). Sage Publications.

Copyright of Managing Sport & Leisure is the property of Routledge and its content may not be copied or emailed to multiple sites or posted to a listserv without the copyright holder's express written permission. However, users may print, download, or email articles for individual use.